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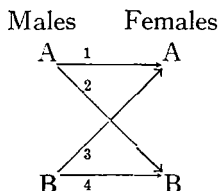
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INTRODUCTION

The recent biological definitions of species by Dobzhansky (1941), Mayr (1942), and others have stressed the attainment of reproductive isolation between two populations as the critical step in speciation. As a result, considerable interest has been aroused in the study of sexual isolation at various levels of divergence. The data, which have been accumulating quite rapidly, have been obtained in various ways, but interpretation of these results on a common basis has been rather difficult. It is desirable, therefore, to have some simple means of comparing the results from the different types of mating experiments.

If the mating behavior or the degree of sexual isolation between two groups is under study, the mating possibilities are of the following four kinds.



With the male type designated first, these possibilities may conveniently be written AA (1) and BB (4) (homogamic matings) and AB (2) and BA (3) (heterogamic matings). Experimentally (and under natural conditions), various mating situations may exist. Each of the four matings may be studied separately, a method generally referred to as "pair matings." In a second experimental situation, males of one type may be placed with females of both types. This kind of experiment, the one most commonly used, may be called "male multiple choice." Similarly, females of one

type may be placed with both types of males, a method which may be called "female multiple choice." Finally, both types of males and both types of females may be put together, probably most nearly approaching natural conditions. This situation perhaps may best be called simply "multiple choice."

Various indices have been proposed for the measurement of sexual isolation (Stalker, 1942; Levene and Dobzhansky, 1944; Bateman, 1949; Levene, 1949), but all have been designed for use with "male multiple choice" experiments. The significance of the simple indices with regard to sexual isolation has often been obscured by "one-sided mating preferences" (Dobzhansky, 1944) or differences in "relative mating propensity" (Bateman, 1949). This difficulty has been somewhat overcome by the use of a joint isolation index (Bateman, 1949) or a coefficient of joint isolation (Levene, 1949). However, the averaging of the two simple indices to give a single joint index in turn obscures the effects of selective mating. Bateman has taken this fact into account by deriving an index of female mating propensity.

Any analysis of data from mating experiments, therefore, should separate the effects of sexual isolation from those of various sorts of selective mating. Two measurements are desirable, one of the degree of sexual isolation, the other of the strength of selection for one type or the other.

SEXUAL ISOLATION

The measurement of sexual isolation involves making an estimate of the genetic isolation of one group from the other. This estimate should take into

account the possibility of gene flow in either or both directions since unidirectional gene flow will prevent the attainment of any degree of genetic diversity between the two groups about as effectively as gene flow in both directions.

A second requirement for an isolation index is that the effects of selective mating be minimized so that only those factors leading to true sexual isolation will affect the size of the figure obtained. Furthermore, applicability of the index to data from the various types of experiments will greatly enhance its usefulness.

A convenient measure of the degree of sexual isolation between two groups may be obtained from the following equation:

$$(1) \quad I = \frac{AB + BA}{AA + BB}.$$

This term may be called the *isolation estimate*. It is designed for use when equal numbers of the types compared are present (for example, equal numbers of each female type in male choice experiments). This situation has been used in practically all experiments. However, since Levene and Dobzhansky (1945) have shown that the ratio of the probability of a heterogamic mating to the probability of a homogamic mating is a fixed constant relatively independent of the proportions of individuals, the value of I obtained in this way would appear to be generally descriptive of the sexual isolation between the groups. It will be seen that I is the ratio of heterogamic matings to homogamic matings. If there is no interbreeding between the groups so that they are completely sexually isolated, the value of I will be zero. On the other hand, if there is no sexual isolation, I will be approximately one. There would be no upper limit to I if both types of heterogamic matings occurred more often than the homogamic matings. However, this situation, which is quite improbable, has not yet been found experimentally so that in practice the value of I will vary from zero for complete isolation to one for random

mating. The statistical significance of values intermediate between zero and one can be readily determined by the usual chi-square test.

The effects of different "mating propensities" by the female types or by the male types are, in effect, cancelled out because both types of females and both types of males appear in both numerator and denominator.

Levene (1949) has pointed out that different durations for reciprocal mating experiments may influence the joint isolation index. His coefficient of joint isolation is designed to minimize this effect. In the calculation of I also, the effect of different exposure times in reciprocal male or female multiple choice experiments is somewhat offset. The male multiple choice experiments may be used to demonstrate this fact. The basis of I in that case is the difference between the female types in frequency of mating. Because the sum of the times of exposure for both female types is equal, and the sum of these periods in the numerator equals that in the denominator, it can be seen that I is relatively insensitive to differences in the duration of the reciprocal experiments. With "multiple choice" tests the time, of course, is the same for all types. The calculation of I from data of pair matings is most susceptible to spurious fluctuations due to different exposure times.

SELECTIVE MATING

For the evaluation of mating data, some measure of the degree of selective mating is also desirable. Bateman (1949), recognizing this need, derived a *mating propensity index* for the special case of male multiple choice experiments. This index measures the relative mating propensities of the females. A similar index for the males would be useful.

Male multiple choice experiments furnish information only about the relative mating frequencies of the females, and female multiple choice experiments only

about those of the males. Indices of both types can be derived from multiple choice tests and from pair matings, but the time factor must be controlled with the latter for significant results.

The relative frequency of mating by the two female types may be determined from the equation:

$$(2) \quad M_f = \frac{AA + BA}{BB + AB}.$$

This relationship may be called the *female mating ratio*. Similarly, the *male mating ratio* is expressed as

$$(3) \quad M_m = \frac{AA + AB}{BB + BA}.$$

These values are also designed to be used with the most common experimental situation in which the numbers of the types being tested are equal. If types A and B differ in a single allele which is responsible for these mating differences, and the value of I indicates that there is no sexual isolation, the mating ratios can be used to estimate the changes in gene frequency in a population due to selective mating. In this case it is assumed that there is a fixed probability that an individual of a given type will mate. The probability of a given mating between a male and a female is then the product of these separate probabilities. This assumption may not be completely valid, but it has been extremely useful in predicting trends due to selective mating in populations.

For the case of an autosomal recessive, with the heterozygote behaving like the homozygous dominant, the following equations may be derived.

Using symbols drawn from Hogben (1946), let

- D_m = dominant males
- D_f = dominant females
- R_m = recessive males
- R_f = recessive females
- H_m = heterozygous males
- H_f = heterozygous females
- $D_m + R_m + H_m = 1.0$
- $D_f + R_f + H_f = 1.0$

Then,

$$(4) \quad R_{n+1} = \frac{(\frac{1}{2}H_{m_n} + M_m R_{m_n}) \times (\frac{1}{2}H_{f_n} + M_f R_{f_n})}{(1 - R_{f_n} + M_f R_{f_n}) \times (1 - R_{m_n} + M_m R_{m_n})},$$

$$(5) \quad D_{n+1} = \frac{(D_{m_n} + \frac{1}{2}H_{m_n})(D_{f_n} + \frac{1}{2}H_{f_n})}{(1 - R_{f_n} + M_f R_{f_n}) \times (1 - R_{m_n} + M_m R_{m_n})},$$

$$(6) \quad H_{n+1} = 1 - R_{n+1} - D_{n+1}.$$

The value for R_{n+1} (similarly for D_{n+1} and H_{n+1}) can be substituted for both R_m and R_f since the gene is autosomal.

The equations are somewhat simpler if there is deviation from random mating in only one sex. If nonrandom mating is found only in the males, for example, the equations become

$$(7) \quad R_{n+1} = \frac{(\frac{1}{2}H_{m_n} + M_m R_{m_n})(\frac{1}{2}H_{f_n} + R_{f_n})}{1 - R_{m_n} + M_m R_{m_n}},$$

and so forth.

For a sex-linked recessive, the proportions of the various male and female genotypes in each generation can be obtained as follows:

Let

- RR = recessive females
- RY = recessive males
- RD = heterozygous females
- DD = dominant females
- DY = dominant males
- $RR + RD + DD = 1.0$
- $RY + RD = 1.0$

Then,

$$(8) \quad RY_{n+1} = \frac{\frac{1}{2}RD_n + M_f RR_n}{1 - RR_n + M_f RR_n},$$

$$(9) \quad DY_{n+1} = 1 - RY_{n+1},$$

$$(10) \quad RR_{n+1} = \frac{M_m RY_n}{DY_n + M_m RY_n} \cdot RY_{n+1},$$

$$(11) \quad DD_{n+1} = \frac{DY_n}{DY_n + M_m RY_n} \cdot DY_{n+1},$$

$$(12) \quad RD_{n+1} = 1 - RR_{n+1} - DD_{n+1}.$$

If only the males are selectively eliminated, a situation which has been found to be quite common experi-

mentally, the equations are as follows:
Let

p_f = frequency of dominant gene in the females,

q_f = frequency of recessive gene in the females,

$p_m = DY$ = frequency of dominant gene in the males,

$q_m = RY$ = frequency of recessive gene in the males.

Then,

$$(13) \quad RY_{n+1} = q_{f_n} = q_{m_{n+1}},$$

$$(14) \quad DY_{n+1} = p_{f_n} = p_{m_{n+1}},$$

$$(15) \quad RR_{n+1} = \frac{M_m q_{m_n} q_{f_n}}{p_{m_n} + M_m q_{m_n}},$$

$$(16) \quad DD_{n+1} = \frac{p_{m_n} p_{f_n}}{p_{m_n} + M_m q_{m_n}},$$

$$(17) \quad RD_{n+1} = 1 - RR_{n+1} - DD_{n+1}$$

$$(18) \quad q_{f_{n+1}} = RR_{n+1} + \frac{1}{2} RD_{n+1},$$

$$(19) \quad p_{f_{n+1}} = 1 - q_{f_{n+1}}.$$

From equation (13) it can be seen that the frequency of recessive males in generation $n + 1$ equals the frequency of the recessive gene in the females of the previous generation, n .

CALCULATION OF I , M_f AND M_m

Table 1 has been drawn up to illustrate the calculation of the three ratios from all four types of mating experiments. The more common results showing sexual isolation and selective mating are demonstrated.

The multiple choice data of Reed and Reed (1950) with white-eyed and wild type *Drosophila melanogaster* show that the cause of the deviations from random mating is the relative inability of the white-eyed males to obtain mates as compared with the wild type.

Mayr and Dobzhansky's work (1945) with *Drosophila pseudoobscura* and *Drosophila persimilis* shows a high degree of sexual isolation. There is a tendency for *pseudoobscura* females to mate more

TABLE 1

Type of experiment	Source	Female type	Male types		I	χ^2	M_f	χ^2	M_m	χ^2
Multiple choice	Reed and Reed (1950) Table 1	white eyed heterozygous	white eyed 90 64	wild type 100 105	0.84	2.68	1.12	1.23	0.75	7.24†
Male multiple choice	Mayr and Dobzhansky (1945) Table 2	<i>pseudoobscura</i> <i>persimilis</i>	<i>pseudoobscura</i> 72/82 8/83	<i>persimilis</i> 27/119 81/127	0.24	70.19†	1.16	1.07	—	—
	Dobzhansky and Streisinger (1944) Table 1	Chilpancingo Huichihuayan	69/72 23/70	Huichihuayan 41/47 15/48	0.78	2.22	2.87	34.37†	—	—
	Patterson, McDonald and Stone (1947) Table 4	<i>americana</i> <i>texana</i>	70/115 36/115	<i>texana</i> 62/112 34/111	0.94	0.16	1.88	18.78†	—	—
	Tan (1946) Table 1	<i>y sn v co sh</i> <i>ast</i>	<i>y sn v co sh</i> 15/44 2/46	<i>ast</i> 26/39 23/38	0.71	1.92	1.66	4.08*	—	—
	Merrell (1949) Table 3	cut heterozygous	cut 15 24	wild type 11 22	0.95	0.06	0.57	5.56*	—	—
	Merrell (1949) Table 2	cut heterozygous	cut 3 6	wild type 23 41	0.66	3.08	—	—	0.14	41.44†
Female multiple choice	Merrell (1949) Table 2	cut heterozygous	cut 3 6	wild type 23 41	0.66	3.08	—	—	0.14	41.44†
Pair mating	†Patterson, Stone and Griffen (1940) Table 1	<i>americana</i> <i>texana</i>	74 28	<i>texana</i> 44 54	0.56	15.68†	1.44	6.48*	1.04	0.08

* Statistically significant.

† Statistically highly significant.

‡ Based on counts of at least 100 vials. χ^2 on that basis.

often than *persimilis* females, but it is not statistically significant in this case.

The results with the Chilpancingo and Huichihuayan strains of *Drosophila prosaltans* show that, although the groups are not sexually isolated, the Chilpancingo females mate much more frequently than Huichihuayan females. Tan's data with mutants of *Drosophila pseudoobscura* and the data on male choice for *cut* in *Drosophila melanogaster* are similar to the *prosaltans* results, one female type mating more often than the other. These similarities may or may not have similar causes. For Tan's data, Bateman's joint isolation index gives a value of 0.37, from which he concludes that "The overall effect of this one-sided action of y on mating behavior is the sexual isolation of y ." Actually, only one type of mating is blocked, that between y *sn v co sh* males and *ast* females, but the exchange of genes is still possible via the reciprocal mating. Therefore, these groups should not be considered sexually isolated to any great extent, and a situation of this kind will never lead to complete sexual isolation.

The value of I for *cut* obtained from female multiple choice experiments differs from that obtained from male choice experiments, but neither is statistically significant. However, M_m shows the notable lack of mating success by the *cut* males, and M_f a lower mating frequency for *cut* females.

The data (1940) on pair matings between *Drosophila americana americana* and *Drosophila americana texana* (see Stone and Patterson, 1947) show that the *americana* females mate somewhat more often than the *texana* females. Furthermore, even though natural hybrids are still found, these subspecies are already to some extent sexually isolated. These results can be compared to the data from male multiple choice experiments of Patterson, McDanald, and Stone (1947). Again the *americana* females mate more frequently than *texana* females, but in this case, no sexual isolation is evident.

This difference is not surprising since Streisinger (1948) demonstrated that *Drosophila* males show relatively little discrimination in mating when given a "choice" of two species of etherized females. Therefore, sexual isolation between closely related groups is evidently less apt to show up in experiments with male choice.

SUMMARY

For the measurement of the sexual isolation between two populations, an *isolation estimate* is derived for use with data from many kinds of mating experiments. This *isolation estimate* is relatively insensitive both to the effects of selective mating and to the effects of different durations of exposure of females to males in reciprocal experiments, thus providing a consistent estimate of the degree of isolation.

A *female mating ratio* and a *male mating ratio* are derived to measure the relative frequency of mating by the different types of the same sex. These ratios are further used in equations for the calculation of the effect of selective mating on gene frequencies when the two types differ at one locus.

Typical examples are chosen from the various kinds of mating experiments to illustrate the calculation and interpretation of these three terms.

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